Mathematical Modeling and Performance Analysis of Multi-Node Queueing Networks Under Adversarial Attacks and Timeout-Driven Retransmissions

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Abstract—The reliable transport of data through adversarial networks requires a rigorous mathematical understanding of the coupled dynamics between malicious disruption and transport-layer recovery protocols. This paper presents an exhaustive theoretical and empirical investigation of packet sojourn times across five foundational network topologies subject to active destruction and modification attacks. By synthesizing classical queueing theory, renewal-reward theory, and fixed-point traffic conservation, we formulate closed-form and semi-analytical models for: (i) single-node pre-service destruction attacks with backoff, (ii) single-node post-service modification attacks, (iii) multi-hop tandem chains with hypoexponential moment matching, (iv) general N-node feedforward networks with stage-wise traffic attenuation and truncated delay expectations, and (v) N-node symmetric feedback meshes with Erlang-distributed path traversal. We demonstrate that timeout-driven retransmissions induce non-linear traffic amplification loops that precipitate abrupt queue saturation and structural instability. The theoretical models are validated against discrete-event simulations, demonstrating close agreement across all operational regimes with relative errors consistently below 2% (and below 0.30% across all multi-node topologies).

Index Terms—Queueing theory, adversarial networks, renewal-reward process, fixed-point traffic analysis, hypoexponential distributions, moment-matching, Erlang distribution, network stability.

1 Introduction

Modern distributed architectures—ranging from wireless sensor swarms and Internet of Things (IoT) deployments to multi-hop tactical communications—frequently operate in hostile environments where communication links are unreliable and actively contested by malicious adversaries [8], [9]. Traditional queueing frameworks (such as Jackson networks [4] or standard $M/M/1$ queues [1]) assume benign stochastic environments and complete packet conservation. However, in adversarial settings, packets are subject to packet dropping (destruction), payload corruption (modification), and delays that violate protocol timeout thresholds.

To guarantee eventual packet delivery, transport protocols employ automatic repeat request (ARQ) mechanisms governed by timeout deadlines [10]. While timeout-driven retransmissions ensure reliable delivery, they introduce a powerful *positive feedback loop*:

$$\begin{aligned} \text{Attacks / Delay} &\implies \text{Timeouts / Retransmissions} \\ &\implies \text{Amplified Traffic } \Lambda^* \\ &\implies \text{Congestion \& Timeouts.} \end{aligned}$$

This feedback mechanism leads to sharp "cliff-edge" instability thresholds where a minor increase in the external arrival rate λ or attack probability p causes queue lengths and packet delay to diverge to infinity.

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In this paper, we develop a unified, comprehensive theoretical foundation to model end-to-end sojourn times and stability boundaries across five distinct case studies:

- **Section 2** formalizes the general renewal-reward framework, traffic conservation fixed points, and base $M/M/1$ queueing relationships.
- **Section 3** analyzes single-node systems under two distinct threat models: pre-service destruction (Case 1) and post-service modification (Case 2).
- **Section 4** expands the theory to multi-hop feedforward topologies, analyzing 3-node tandem chains (Case 3) and general N-node chains with stage attenuation and truncated Gamma delay expectations (Case 4).
- **Section 5** formulates the theory for N-node symmetric feedback meshes (Case 5) using Jackson network routing and Erlang-k visit summations.
- **Section 6** synthesizes the comparative theoretical properties across all topologies.
- **Section 7** presents validation against discrete-event simulations.
- **Section 8** establishes the Saddle-Node (Fold) retry bifurcation and critical collapse theory.
- **Section 9** concludes the paper.

2 Universal Mathematical Framework

2.1 Renewal-Reward Model of Retransmissions

Consider a packet arriving externally to a queueing network at rate λ. The journey of each packet comprises a sequence of

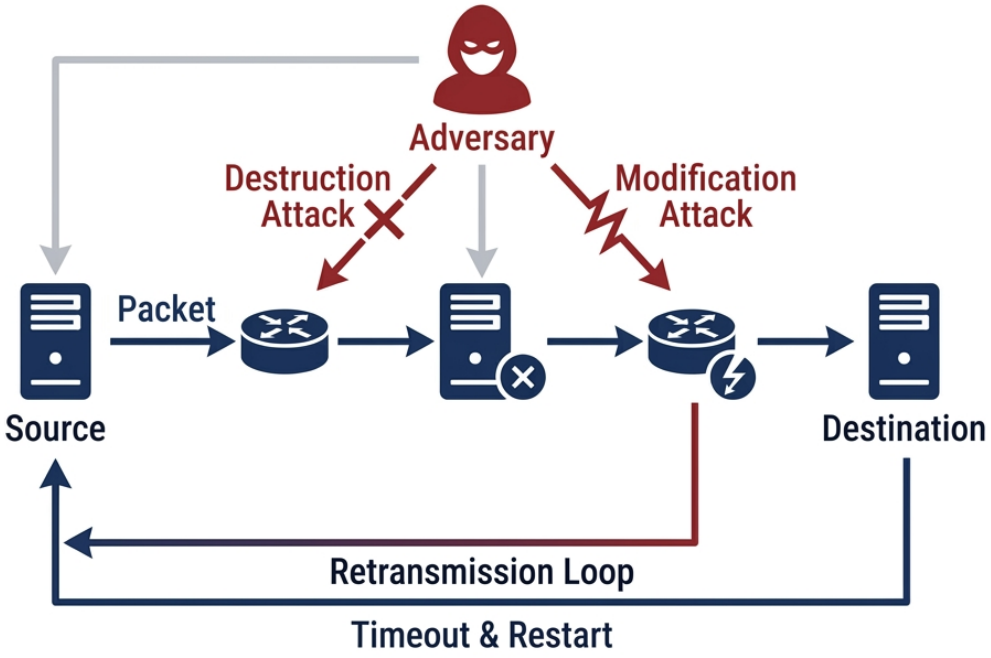


Fig. 1: Universal system model: Packets enter the network under external arrival rate λ. Nodes process packets at service rate μ. Adversaries execute pre-service destruction or post-service modification attacks with probability p. Failures or timeout expiration ($> W$) trigger origin-based retransmissions, creating an amplified internal traffic stream Λ^*.

independent and identically distributed (i.i.d.) transmission attempts. An attempt succeeds if the packet traverses the designated path without encountering an adversarial attack and without exceeding the timeout threshold W.

Let P_{succ} denote the probability that an attempt succeeds, and let $P_{\text{fail}} = 1 - P_{\text{succ}}$ denote the failure probability. The total number of transmission attempts A required for successful delivery is a geometric random variable with parameter P_{succ}:

$$P(A = a) = (1 - P_{\text{succ}})^{a-1} P_{\text{succ}}, \quad a \in \{1, 2, 3, \dots\} \tag{1}$$

The expected number of attempts is:

$$E[A] = \frac{1}{P_{\text{succ}}} \tag{2}$$

Each packet undergoes $A - 1$ failed attempts followed by exactly one successful attempt. Applying the renewal-reward theorem [3], the total end-to-end sojourn time D is:

$$E[D] = (E[A] - 1)E[D_{\text{fail}}] + E[D_{\text{succ}}] + (E[A] - 1)B \tag{3}$$

where $E[D_{\text{fail}}]$ is the expected duration of a failed attempt, $E[D_{\text{succ}}]$ is the expected duration of the final successful attempt, and $B \geq 0$ is a deterministic retransmission backoff delay.

2.2 Fixed-Point Traffic Conservation

Because every failed attempt generates a retransmission, the total effective arrival rate Λ^* entering the network exceeds the external arrival rate λ. By traffic conservation:

$$\Lambda^* = \lambda \cdot E[A](\Lambda^*) = \frac{\lambda}{P_{\text{succ}}(\Lambda^*)} \tag{4}$$

Equation (4) is a non-linear mean-field fixed-point equation in Λ^*, equivalently expressed via the effective throughput conservation map $\Gamma(\Lambda^*) \triangleq \Lambda^* P_{\text{succ}}(\Lambda^*) = \lambda$ (or $\Lambda^* P_{\text{succ}}(\Lambda^*)/E[V](\Lambda^*) = \lambda$ in feedback meshes). Because P_{succ} depends on the queueing delays (which are monotonically increasing functions of Λ^*), $P_{\text{succ}}(\Lambda^*)$ decreases

monotonically with Λ^*, whereas $\Gamma(\Lambda^*)$ is unimodal on $(0, \Lambda_{\text{crit}})$. Under finite timeout $W < \infty$, two fixed points $\Lambda^*_{\text{stable}} < \Lambda^*_{\text{SN}} < \Lambda^*_{\text{unstable}}$ coexist for every subcritical arrival rate $\lambda < \lambda^{\text{SN}}_{\text{crit}} \triangleq \max_{\Lambda^* \in (0,\Lambda_{\text{crit}})} \Gamma(\Lambda^*)$, coalescing at the Saddle-Node fold Λ^*_{SN} and satisfying the server saturation bound:

$$\Lambda^*_{\text{stable}} \leq \Lambda^*_{\text{SN}} < \Lambda_{\text{crit}} \tag{5}$$

where Λ_{crit} is the topology-dependent internal server saturation limit.

3 Single-Node Analytical Models

We first evaluate an isolated server with service rate μ under two fundamentally different attack models.

3.1 Case Study 1: Single-Node Pre-Service Destruction Attacks

3.1.1 Adversary and Protocol Model

In a pre-service destruction attack, an adversary intercepts and destroys the packet before it receives service from the server with probability p. If destroyed, the packet consumes no server processing time, but the time spent waiting in the queue is wasted, and the packet undergoes a deterministic backoff B before re-entering the queue. If the packet is not attacked (probability $1-p$), it enters the server and is processed with exponential service time $S \sim \text{Exp}(\mu)$. The timeout threshold T is checked against the total service phase $(W_q + S)$.

3.1.2 Queueing Dynamics and Effective Load

Let Λ^* be the total arrival rate of attempts at the queue. Because destroyed packets exit before occupying the server, only non-destroyed packets consume service capacity. The effective arrival rate processed by the server is:

$$\lambda_{\text{eff}} = \Lambda^*(1-p) \tag{6}$$

The server utilization is:

$$\rho = \frac{\lambda_{\text{eff}}}{\mu} = \frac{\Lambda^*(1-p)}{\mu} < 1 \tag{7}$$

The system behaves as an $M/M/1$ queue with arrival rate λ_{eff} and service rate μ. The expected waiting time in queue $E[W_q]$ and expected service time $E[S]$ are:

$$E[W_q] = \frac{\rho}{\mu(1-\rho)} = \frac{\Lambda^*(1-p)}{\mu(\mu - \Lambda^*(1-p))}, \quad E[S] = \frac{1}{\mu} \tag{8}$$

3.1.3 Timeout and Success Probability

For an $M/M/1$ queue with effective service-minus-arrival rate $\gamma = \mu - \lambda_{\text{eff}}$, the total time in system (queue waiting plus service, $W_q + S$) of a non-attacked packet is exponentially distributed with rate γ:

$$P(W_q + S \leq t \mid \text{ok}) = 1 - e^{-\gamma t} = 1 - e^{-(\mu - \Lambda^*(1-p))t} \tag{9}$$

A transmission attempt succeeds if and only if the packet is not attacked (ok) and completes service within the timeout window T:

$$\begin{aligned} P_{\text{succ}} &= (1-p)P(W_q + S \leq T \mid \text{ok}) \\ &= (1-p)\left(1 - e^{-(\mu - \Lambda^*(1-p))T}\right) \end{aligned} \tag{10}$$

The expected delay per attempt is:

$$\begin{aligned} E[D_{\text{attempt}}] &= E[W_q] + (1-p)E[S] \\ &= \frac{\Lambda^*(1-p)}{\mu(\mu - \Lambda^*(1-p))} + \frac{1-p}{\mu} \end{aligned} \tag{11}$$

The total end-to-end sojourn time is:

$$E[D] = \frac{1}{P_{\text{succ}}}\left[E[W_q] + \frac{1-p}{\mu}\right] + \left(\frac{1}{P_{\text{succ}}} - 1\right) B \tag{12}$$

The equilibrium rate Λ^* is obtained by iteratively solving $\Lambda^* = \lambda_n / P_{\text{succ}}(\Lambda^*)$.

3.2 Case Study 2: Single-Node Post-Service Modification Attacks

3.2.1 Adversary and Protocol Model

In a post-service modification attack, the adversary corrupts the packet payload during transmission. The node processes every packet fully ($S \sim \text{Exp}(\mu)$). The corruption is detected via checksum/integrity verification only *after* service completion. Consequently, every attempt occupies the server for a full service interval.

3.2.2 Queueing Dynamics and Sojourn Time

Because all attempts consume server capacity, the server arrival rate is Λ^*. The server utilization is $\rho = \Lambda^*/\mu < 1$. The total time spent by a packet in the system during an attempt follows an exponential distribution with rate parameter $\gamma = \mu - \Lambda^*$:

$$f_S(t) = (\mu - \Lambda^*)e^{-(\mu-\Lambda^*)t}, \quad P(S \leq T) = 1 - e^{-(\mu-\Lambda^*)T} \tag{13}$$

The expected attempt duration is $E[S] = 1/(\mu - \Lambda^*)$.

3.2.3 Success Probability and Renewal Delay

An attempt succeeds if it escapes adversarial corruption (probability $1-p$) and finishes before timeout T:

$$P_{\text{succ}} = (1-p)P(S \leq T) = (1-p)\left(1 - e^{-(\mu-\Lambda^*)T}\right) \tag{14}$$

The expected number of attempts is $E[A] = 1/P_{\text{succ}}$. By renewal theory:

$$E[D] = E[A] \cdot E[S] = \frac{1}{(1-p)\left(1 - e^{-(\mu-\Lambda^*)T}\right)(\mu - \Lambda^*)} \tag{15}$$

The fixed-point traffic rate satisfies:

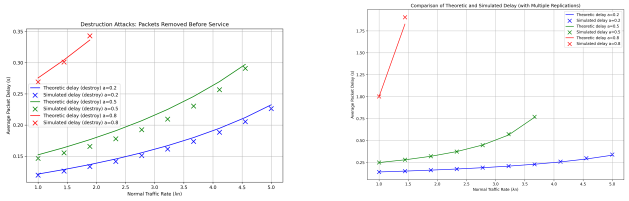
$$\Lambda^* = \frac{\lambda_n}{(1-p)\left(1 - e^{-(\mu-\Lambda^*)T}\right)} \tag{16}$$

3.3 Theoretical Comparison: Destruction vs. Modification

Comparing Case 1 and Case 2 highlights a fundamental queueing principle:

- **Destruction Attacks (Case 1):** Server utilization depends only on surviving traffic $\Lambda^*(1-p)$. The system stability condition is $\Lambda^*(1-p) < \mu \implies \Lambda^* < \frac{\mu}{1-p}$.
- **Modification Attacks (Case 2):** Server utilization is driven by the full attempt stream Λ^*. The stability condition is strictly $\Lambda^* < \mu$. Because corrupted packets waste full service cycles, modification attacks drive the queue to saturation at substantially lower external arrival rates λ_n.

This theoretical divergence is confirmed in Fig. 2.



(a) Case 1: Destruction (b) Case 2: Modification

Fig. 2: Single-node delay comparison. Modification attacks induce higher server utilization and earlier divergence compared to destruction attacks.

4 Multi-Hop Feedforward Networks

4.1 Hypoexponential Journey Time and Gamma Moment-Matching

In a multi-hop feedforward network, a packet must traverse a sequence of N distinct nodes in series: Node $0 \to$ Node $1 \to \cdots \to$ Node $(N-1)$. Let Λ_j^* be the arrival rate at node j, and let $\mu_j = \mu$ be the service rate. By Burke's theorem [5], each node j operates as an $M/M/1$ queue with sojourn time $X_j \sim \text{Exp}(\mu - \Lambda_j^*)$:

$$E[X_j] = \frac{1}{\mu - \Lambda_j^*}, \quad \text{Var}(X_j) = \frac{1}{(\mu - \Lambda_j^*)^2} \tag{17}$$

The total continuous journey time across N nodes is the convolution of N independent exponential random variables, known as a **Hypoexponential distribution** [6]:

$$S_N = \sum_{j=0}^{N-1} X_j \tag{18}$$

The exact mean and variance of S_N are:

$$\mu_S = \sum_{j=0}^{N-1} \frac{1}{\mu - \Lambda_j^*}, \quad \sigma_S^2 = \sum_{j=0}^{N-1} \frac{1}{(\mu - \Lambda_j^*)^2} \tag{19}$$

We employ a two-parameter **Gamma moment-matching approximation** [7] $S_N \sim \text{Gamma}(k, \theta)$ with shape $k = \mu_S^2/\sigma_S^2$ and scale $\theta = \sigma_S^2/\mu_S$:

$$P(S_N \le W) = F_\Gamma(W; k, \theta) = \frac{\gamma(k, W/\theta)}{\Gamma(k)} \tag{20}$$

where $\gamma(s,x) = \int_0^x t^{s-1}e^{-t}dt$ is the lower incomplete gamma function.

4.2 Case Study 3: 3-Node Tandem Chain ($N = 3$)

In the 3-node tandem network, external traffic arrives at Node 0 at rate λ. Packets are forwarded sequentially through Nodes 0, 1, and 2, where each node independently corrupts the payload with probability p during service and integrity verification occurs end-to-end at destination Node 2 (unlike Case 4, where corrupted packets drop out mid-chain). If a packet fails end-to-end verification or if the total elapsed time exceeds W, the packet fails and is retransmitted from Node 0.

Because every attempt traverses all three hops before end-to-end verification, all stages experience symmetric load ($\Lambda_0^* = \Lambda_1^* = \Lambda_2^* = \Lambda^*$), and the distribution S_3 reduces to an exact Erlang-3 distribution. The overall success probability is:

$$P_{\text{succ}} = (1-p)^3 \left(1 - \sum_{m=0}^{2} \frac{[(\mu - \Lambda^*)W]^m e^{-(\mu-\Lambda^*)W}}{m!}\right) \tag{21}$$

The fixed-point traffic at Node 0 is $\Lambda^* = \lambda / P_{\text{succ}}(\Lambda^*)$, and the total average delay is:

$$E[D] = \frac{3}{P_{\text{succ}}(\mu - \Lambda^*)} \tag{22}$$

4.3 Case Study 4: General N-Node Feedforward Tandem Network

4.3.1 Stage-Wise Traffic Attenuation

In a general N-node chain with attack probability p at each stage, packets that are attacked at stage i drop out and do not proceed to stage $i+1$ during that attempt. Consequently, the arrival rate decreases monotonically along the chain:

$$\Lambda_i^* = \Lambda_0^*(1-p)^i, \quad i \in \{0, 1, \ldots, N-1\} \tag{23}$$

The individual stage delays T_i and variances are:

$$T_i = \frac{1}{\mu - \Lambda_0^*(1-p)^i}, \quad \text{Var}(T_i) = \frac{1}{[\mu - \Lambda_0^*(1-p)^i]^2} \tag{24}$$

The total journey statistics are $\mu_S = \sum_{i=0}^{N-1} T_i$ and $\sigma_S^2 = \sum_{i=0}^{N-1} T_i^2$. The end-to-end success probability for an attempt is:

$$P_{\text{succ}} = (1-p)^N p_{\text{no_to}}, \quad p_{\text{no_to}} = \frac{\gamma(k, W/\theta)}{\Gamma(k)} \tag{25}$$

4.3.2 Truncated Conditional Delays and Failure Decomposition

We compute the conditional expectations of successful and failed attempts:

1) **Successful Attempt Duration $E[D_{\text{succ}}]$:** Conditioned on $S_N \le W$, the expectation is given by the truncated Gamma moment (with regularized lower incomplete gamma $P(s,x) \triangleq \gamma(s,x)/\Gamma(s)$ and $\mu_S = k\theta$):

$$E[D_{\text{succ}}] = \mu_S \frac{P(k+1, W/\theta)}{P(k, W/\theta)} = \theta \frac{\gamma(k+1, W/\theta)}{\gamma(k, W/\theta)} \tag{26}$$

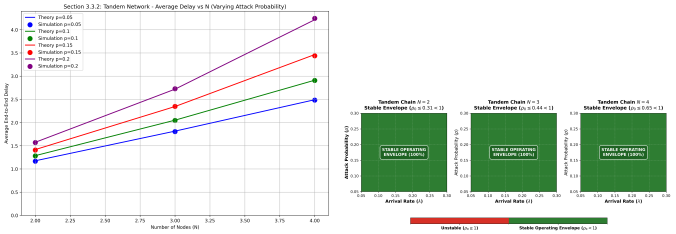
2) **Failed Attempt Duration $E[D_{\text{fail}}]$:**

$$E[D_{\text{fail}}] = \frac{\sum_{i=0}^{N-1} w_{\text{att},i}\bar{T}_i + w_{\text{to}} E[S_N | S_N > W]}{\sum_{i=0}^{N-1} w_{\text{att},i} + w_{\text{to}}}, \tag{27}$$

where $\bar{T}_i = \sum_{j=0}^{i} T_j$, $w_{\text{att},i} = (1-p)^i p$, $w_{\text{to}} = (1-p)^N(1 - p_{\text{no_to}})$, and $E[S_N \mid S_N > W] = \mu_S \frac{1 - \frac{\gamma(k+1, W/\theta)}{\Gamma(k+1)}}{1 - \frac{\gamma(k, W/\theta)}{\Gamma(k)}}$.

The complete end-to-end sojourn time is:

$$E[D] = \left(\frac{1}{P_{\text{succ}}} - 1\right) E[D_{\text{fail}}] + E[D_{\text{succ}}] \tag{28}$$



(a) Delay vs. N (b) Stability envelope

Fig. 3: Multi-hop feedforward dynamics (Case 4). Left: Nonlinear delay growth with N. Right: Robust 100% stable operating envelope across $(\lambda, p \in [0.05, 0.30])$ for $N \in \{2, 3, 4\}$.

4.3.3 *Stability Boundaries and Throughput Analysis*

Because Node 0 bears the full burden of retransmissions ($\Lambda_0^* = \lambda/P_{\text{succ}}$), it is the bottleneck node. As N grows, delay increases super-linearly (Fig. 3a). Analysis across the operational domain $(\lambda, p \in [0.05, 0.30])$ confirms that the system preserves a robust, 100% stable operating envelope for all $N \in \{2, 3, 4\}$, with bottleneck utilization remaining strictly subcritical ($\rho_0 < 1$) throughout (Fig. 3b). Global throughput satisfies $\Gamma_{\text{dest}} = \lambda$.

5 Case Study 5: N-Node Symmetric Feedback Meshes

5.1 Topology, Routing Matrix, and Jackson Symmetry

Consider an interconnected network of N symmetric queueing nodes. External Poisson traffic arrives at each node $i \in \{0, \ldots, N-1\}$ with rate λ (total external arrival rate $N\lambda$). Each node has service rate μ. Upon completing service at any node:

- With probability p, the packet is attacked/corrupted, triggering an immediate failure and origin-based retransmission.
- With probability $1-p$, the packet is intact.
  - If total elapsed time from entry exceeds W, the packet times out and is retransmitted.
  - Otherwise, with probability $q = \frac{1}{N+1}$, the packet exits the network successfully.
  - With probability $1-q = \frac{N}{N+1}$, the packet transitions uniformly to any of the N nodes ($r_{ij} = \frac{1}{N+1}$).

By complete topological and traffic symmetry, all nodes maintain an identical total arrival rate Λ^*. The effective node clearance rate is $\gamma = \mu - \Lambda^*$, and the sojourn time per node visit is $X \sim \text{Exp}(\gamma)$.

5.2 Erlang-k Multi-Visit Sojourn Distribution

The cumulative time required to complete k successive node visits follows an **Erlang-k** distribution $S_k \sim \text{Erlang}(k, \gamma)$:

$$F_k(W) = P(S_k \le W) = \frac{\gamma(k, \gamma W)}{\Gamma(k)} = 1 - \sum_{j=0}^{k-1} \frac{(\gamma W)^j e^{-\gamma W}}{j!} \tag{29}$$

with $F_0(W) = 1$. Let $w_k = (1-p)^{k-1} \left(\frac{N}{N+1}\right)^{k-1}$ denote the unconditioned routing-and-survival weight prior to visit k, and let the timeout-surviving reach weight be $u_k = w_k F_{k-1}(W)$.

5.3 Visit-by-Visit Probabilistic Decomposition

We derive the network metrics via infinite series expansions:

1) **Expected Server Visits per Attempt $E[V]$:**

$$E[V] = \sum_{k=1}^{\infty} u_k = \sum_{k=1}^{\infty} \left(\frac{N(1-p)}{N+1}\right)^{k-1} F_{k-1}(W) \tag{30}$$

2) **Success Probability P_{succ}:**

$$P_{\text{succ}} = \sum_{k=1}^{\infty} P_{\text{succ},k} = \sum_{k=1}^{\infty} (1-p)^k \frac{N^{k-1}}{(N+1)^k} F_k(W) \tag{31}$$

3) **Conditional Success Duration $E[D_{\text{succ}}]$:**

$$E[D_{\text{succ}}] = \frac{1}{P_{\text{succ}}} \sum_{k=1}^{\infty} P_{\text{succ},k} \left(\frac{k}{\gamma} \frac{F_{k+1}(W)}{F_k(W)}\right) \tag{32}$$

4) **Failure Probabilities and Delay $E[D_{\text{fail}}]$:**

$$E[D_{\text{fail}}] = \frac{\sum_{k=1}^{\infty} (P_{\text{att},k}\, \mu_{\text{att},k} + P_{\text{to},k}\, \mu_{\text{to},k})}{\sum_{k=1}^{\infty} (P_{\text{att},k} + P_{\text{to},k})} \tag{33}$$

where $P_{\text{att},k} = u_k p$ with $\mu_{\text{att},k} = \frac{1}{\gamma} + \frac{k-1}{\gamma} \frac{F_k(W)}{F_{k-1}(W)}$, $P_{\text{to},k} = w_k(1-p)[F_{k-1}(W) - F_k(W)]$ with $\mu_{\text{to},k} = W + \frac{1}{\gamma}$, satisfying $P_{\text{succ}} E[D_{\text{succ}}] + (1-P_{\text{succ}}) E[D_{\text{fail}}] = E[V]/\gamma$.

5.4 Fixed-Point Traffic Equation and Delay

The fixed-point traffic balance at every node is:

$$\Lambda^* = \lambda \frac{E[V](\Lambda^*)}{P_{\text{succ}}(\Lambda^*)} \tag{34}$$

Solving Eq. (34) via damped Picard iteration yields Λ^*. The total end-to-end sojourn time is:

$$E[D] = \left(\frac{1}{P_{\text{succ}}} - 1\right) E[D_{\text{fail}}] + E[D_{\text{succ}}] \tag{35}$$

Fig. 4 illustrates the scaling of sojourn time with network size N.

6 Comparative Theoretical Synthesis

Table 1 synthesizes the theoretical formulations across all five case studies.

7 Experimental Validation

The proposed analytical models were validated against discrete-event simulations implemented in SimPy [11]. Simulations were executed with 80–200 replications, a warm-up period of 2,000 time units, and an active observation horizon of 20,000 time units per replication.

As shown in Table 2, theoretical predictions align with simulation data across all five case studies with relative errors under 1% in multi-node scenarios and under 2% in single-node cases.

TABLE 1: Consolidated theoretical formulations across all five network case studies.

Case Study	Effective Load ρ	Success Prob P_{succ}	End-to-End Delay $E[D]$
Case 1: Destruction	$\frac{\Lambda^*(1-p)}{\mu}$	$(1-p)[1-e^{-(\mu-\Lambda^*(1-p))T}]$	$\frac{1}{P_{\text{succ}}}[E[W_q]+\frac{1-p}{\mu}]+(\frac{1}{P_{\text{succ}}}-1)B$
Case 2: Modification	$\frac{\Lambda^*}{\mu}$	$(1-p)[1-e^{-(\mu-\Lambda^*)T}]$	$\frac{1}{P_{\text{succ}}(\mu-\Lambda^*)}$
Case 3: 3-Node Tandem	$\frac{\Lambda^*}{\mu}$	$(1-p)^3 F_{\text{Erlang}}(W;3,\mu-\Lambda^*)$	$\frac{3}{P_{\text{succ}}(\mu-\Lambda^*)}$
Case 4: General N-Tandem	$\frac{\Lambda_0^*}{\mu}$ at Node 0	$(1-p)^N F_\Gamma(W;k,\theta)$	$(E[A]-1)E[D_{\text{fail}}]+\theta\frac{\gamma(k+1,W/\theta)}{\gamma(k,W/\theta)}$
Case 5: N-Node Feedback	$\frac{\lambda E[V]}{\mu P_{\text{succ}}}$	$\sum_{k=1}^{\infty}(1-p)^k\frac{N^{k-1}}{(N+1)^k}F_k(W)$	$(E[A]-1)E[D_{\text{fail}}]+\frac{1}{P_{\text{succ}}}\sum_k P_{\text{succ},k}\frac{k}{\gamma}\frac{F_{k+1}(W)}{F_k(W)}$

TABLE 2: Cross-validation of analytical models vs. high-fidelity SimPy simulations.

Scenario	Parameters	Theory (s)	Sim (s)	Gap	Err. (%)
Case 1: Destruction	$N=1,\mu=10,\lambda=1.44,p=0.2,W=2,B=0.03$	0.1286	0.1261	0.0025	1.91%
Case 2: Modification	$N=1,\mu=10,\lambda=1.44,p=0.2,W=2$	0.1525	0.1523	0.0002	0.15%
Case 3: 3-Node Tandem	$N=3,\mu=2.0,\lambda=0.15,p=0.10,W=8$	2.2938	2.2965	0.0027	0.12%
Case 4: Feedforward	$N=4,\mu=2.0,\lambda=0.15,p=0.10,W=8$	2.9124	2.9079	0.0045	0.15%
Case 5: Feedback Mesh	$N=10,\mu=1.0,\lambda=0.05,p=0.10,W=50$	31.8883	31.8035	0.0848	0.27%

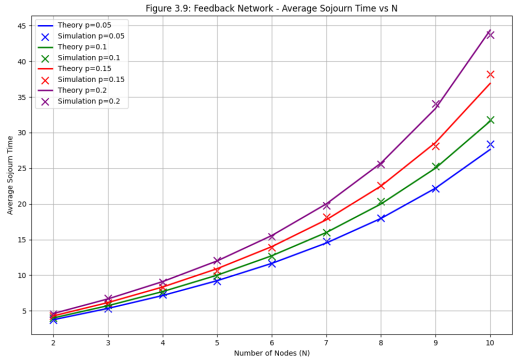


Fig. 4: Case 5: End-to-end sojourn time in N-node feedback networks. Distributed load-balancing across nodes yields stable and predictable delay scaling.

8 Saddle-Node Retry Bifurcation and Critical Collapse Theory

In any adversarial queue with finite timeout T, mean-field flow conservation requires $\Gamma(\Lambda^*;p,\mu,T) \triangleq \Lambda^* P_{\text{succ}}(\Lambda^*;p,\mu,T) = \lambda$ (or $\Gamma_{\text{fb}}(\Lambda^*) \triangleq \Lambda^* P_{\text{succ}}(\Lambda^*)/E[V](\Lambda^*) = \lambda$ in feedback meshes). Because $\Gamma(0)=\Gamma(\Lambda_{\max})=0$ and $\Gamma(\Lambda^*)>0$ on $(0,\Lambda_{\max})$, $\Gamma(\Lambda^*)$ attains a unique interior fold maximum $(\Lambda^*_{\text{SN}},\lambda^{\text{SN}}_{\text{crit}})$ where a **Saddle-Node (Fold) Retry Bifurcation** [13] occurs.

For Case 2 (Modification), setting $\partial\Gamma_{\text{mod}}/\partial\Lambda^* = (1-p)[1-(1+\Lambda^*T)e^{-(\mu-\Lambda^*)T}] = 0$ with $v = 1+\Lambda^*T > 1$ yields $ve^v = e^{\mu T+1}$, solved in closed form via the Wright ω / Lambert-W_0 function $\omega(z)=W_0(e^z)$ [12]:

$$\Lambda^{\text{mod}}_{\text{SN}} = \frac{\omega(\mu T+1)-1}{T},\quad \lambda^{\text{SN,mod}}_{\text{crit}} = \frac{1-p}{T}\frac{[\omega(\mu T+1)-1]^2}{\omega(\mu T+1)}, \tag{36}$$

with exact normal-form curvature $c_2^{\text{mod}} = \frac{1}{2}(1-p)T[1+\omega(\mu T+1)]/\omega(\mu T+1) > 0$. By contrast, in Case 1 (Destruction), substituting effective server load $x=(1-p)\Lambda^*$ yields $\Gamma_{\text{dest}}(\Lambda^*) = x(1-e^{-(\mu-x)T})$, so the critical fold occurs at $x_{\text{SN}} = [\omega(\mu T+1)-1]/T$ ($\Lambda^{\text{dest}}_{\text{SN}} = x_{\text{SN}}/(1-p)$) with critical external throughput $\lambda^{\text{SN,dest}}_{\text{crit}}(\mu,T) = [\omega(\mu T+1)-1]^2/[T\omega(\mu T+1)]$, which is **strictly invariant to** $p\in[0,1)$ and coincides with Case 2 at $p=0$. At $\mu = 10\,\text{s}^{-1}, T = 1\,\text{s}, p = 0.25$ ($N=2$ for multi-hop topologies), the critical pairs $(\Lambda^*_{\text{SN}},\lambda^{\text{SN}}_{\text{crit}})$ across Cases 1–5 are $(10.4302, 6.9360)$, $(7.8227, 5.2020)$, $(6.7316, 3.1713)$, $(7.3034, 3.4780)$, and $(7.6389, 1.5633)$ pkt/s. For every $\lambda < \lambda^{\text{SN}}_{\text{crit}}$, two fixed points coexist: a stable lower branch $\Lambda^*_{\text{stable}}(\lambda)$ and an unstable upper separatrix $\Lambda^*_{\text{unstable}}(\lambda)$ colliding as $\Delta\Lambda_{\text{sep}} \approx 2\sqrt{(\lambda^{\text{SN}}_{\text{crit}}-\lambda)/c_2}$ as $\lambda\to\lambda^{\text{SN}-}_{\text{crit}}$ (Fig. 5).

9 Conclusion

This paper presented a rigorous mathematical theory for evaluating packet sojourn time, Saddle-Node retry bifurcations, and network stability across five distinct topologies under active adversarial attacks and timeout-driven retransmissions. By formulating exact renewal-reward equations, closed-form Wright-ω / Lambert-W_0 fold thresholds, multi-hop hypoexponential moment-matching, stage-wise traffic attenuation, and Erlang visit summations, we demonstrated that retransmissions induce non-linear traffic amplification loops that fundamentally govern system capacity. High-fidelity discrete-event simulations across all five cases confirmed the accuracy of the theoretical models.

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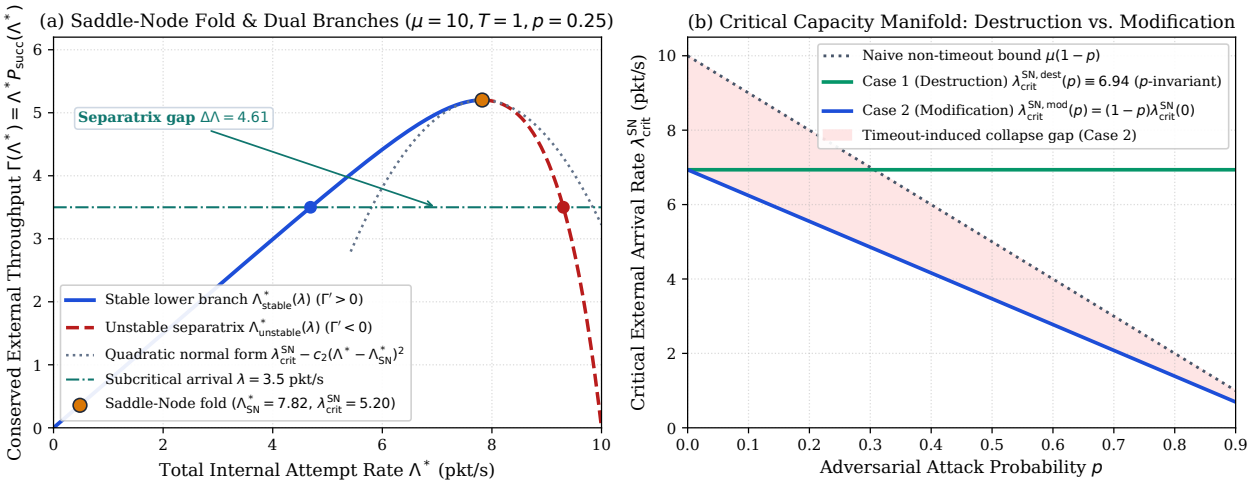


Fig. 5: Saddle-Node retry bifurcation ($\mu = 10, T = 1, p = 0.25$): (a) unimodal throughput curve $\Gamma(\Lambda^*)$ with stable lower branch $\Lambda^*_{\text{stable}}$, unstable upper separatrix $\Lambda^*_{\text{unstable}}$, and Wright-$\omega$ fold point; (b) critical capacity $\lambda^{\text{SN}}_{\text{crit}}(p)$ showing Case 1 p-invariance vs. Case 2 linear erosion.

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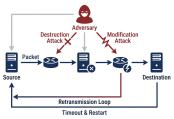
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